

Pulse Width Modulation (PWM) for Motor Control

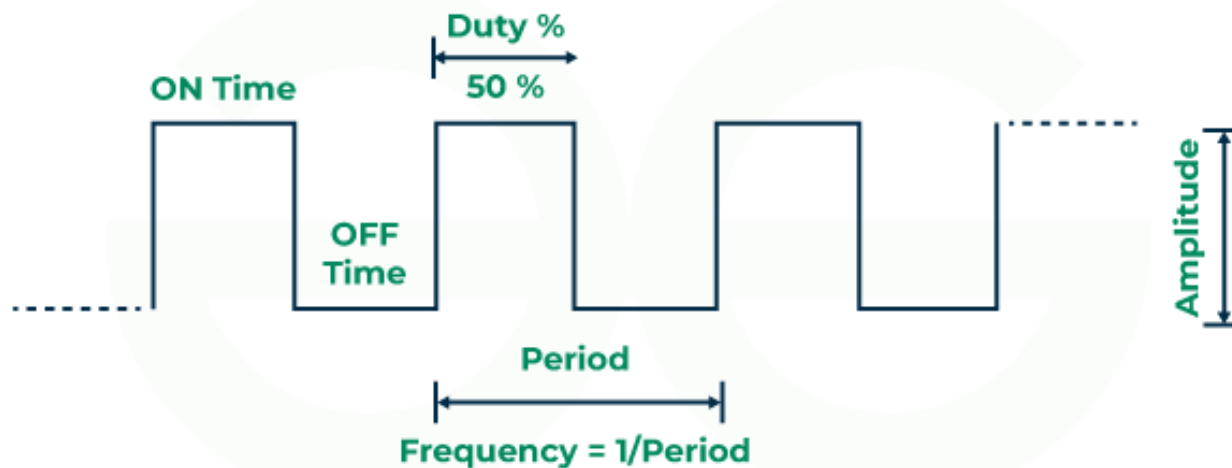
By Bilal Hussain & Nikunj Patel

Frequency, Duty
Cycle, and Slew Rate

Introduction to PWM

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- Pulse Width Modulation (PWM) is a technique to control motor speed and torque.
 - It works by switching the motor's power supply ON and OFF rapidly.
 - The ratio of ON to OFF time determines the motor's effective power.

Picture demonstration for better understanding



Frequency and Time Period

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- Frequency is the number of PWM cycles per second (measured in Hz).
 - Time Period is the reciprocal of frequency ($T = 1/f$).
 - Higher frequencies result in smoother motor operation.
 - Typical motor control uses frequencies in the kHz range.

Duty Cycle

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- Duty Cycle is the percentage of ON time in one PWM cycle.
 - Formula: $\text{Duty Cycle (\%)} = (\text{ON Time} / \text{Total Period}) \times 100$.
 - A higher duty cycle means more average voltage applied to the motor.
 - Controls motor speed: 0% = OFF, 100% = Full Power.

Slew Rate



- Slew Rate is the rate of change of PWM output voltage.
- Determines how quickly the signal transitions between ON and OFF states.
- High slew rate improves response but may cause EMI (Electromagnetic Interference).
- Controlled slew rate reduces noise and improves motor lifetime.

Applications of PWM in Motor Control



Speed control of DC motors.



Torque control in electric vehicles.



Robotics and automation.



Industrial motor drives.

Summary



- PWM controls motor speed and torque by varying ON/OFF switching.
- Frequency and Time Period affect smoothness of control.
- Duty Cycle regulates effective motor voltage.
- Slew Rate impacts signal quality and motor efficiency.



Thank You for
watching